mapped peripherals, including global timer, watchdog and private timer for each Cortex-A9 processor is present in the cluster.

An integrated interrupt controller is an implementation of the generic interrupt controller architecture. Its registers are in the private memory region of Cortex-A9 MPCore processor.

The SCU connects ARM Cortex processors to the memory system through AXI interfaces. It is an important element in maintaining cache coherency-related issues. ARM processors have a shared L2 cache and independent L1 caches. When working in multi-core mode, coherency becomes an important issue to address. The SCU is clocked synchronously and at the same frequency as the processors.

Functions of the SCU are:

- Maintain data cache coherency between Cortex-A9 processors
- Initiate L2 AXI memory accesses
- Arbitrate between Cortex-A9 processors requesting L2 accesses
- Manage ACP accesses

The SCU on Cortex A9 does not provide hardware management of coherency on the instruction cache.

Generic interrupt controller (GIC). The processing system includes a GIC that manages and assigns the interrupts that happen on the system. The interrupt controller is a single functional unit located in a Cortex-A9 multi-processor design. There is one Cortex-A9 processor interface per Cortex-A9 processor. Processors access it by using a private interface through the SCU.

The GIC provides the following:

- Registers for managing interrupt sources, interrupt behaviour and interrupt routing to one or more processors
- Support for ARM architecture security extensions, ARM architecture virtualisation extensions
- Enabling, disabling and generating processor interrupts from hardware (peripheral) interrupt sources

GigaX API for Zynq FPGA

Real-time processing of gigabit Ethernet data in programmable logic is a time-consuming challenge that requires optimised drivers and interface management. SoC devices provide hardware resources that allow this communication, but these lack general-purpose software applications needed for implementation.

GigaX is a lwIP-based API for Xilinx Zynq SoC that establishes a high-speed communication channel between GigaE processing system port and programmable logic. Running in one of the Zynq ARM cores, GigaX processes network and transport headers, and manages SDRAM, Ethernet DMA and AXI interfaces to set up a robust full-duplex data link through the processing system at 200Mbps.

Software API also implements IP filtering and TCP/UDP headers management to allow using the device as an Ethernet bridge, programmable network node, hardware accelerator or network offloader. GigaX is easy to install as a library in SDK project. It can control GigaE peripheral and AXI DMA interface to enable direct communication between Ethernet and VIVADO IP cores.

Based on the open source lwIP stack, GigaX controls Zynq GigaE peripheral to send and receive Ethernet frames to and from the SDRAM via AMBA interconnect bus. After performing IP filtering, Ethernet data is sent to PL through a high performance AXI port, which is also used to send processed data back to the processing system.

Processing system-programmable logic transfer uses AXI DMA implemented in the programmable logic and controlled by GigaX through a general-purpose AXI4-Lite port. Network headers can be kept or removed before reaching the programmable logic. GigaX also allows generation or modification of IP headers of the data received from the programmable logic. Complete full-duplex communication based on DMA and AXI ports allows high-speed data transfer without overloading the ARM processor.

AXI4-Stream interfaces are used to transfer Ethernet data to and from IP cores. GigaX data caching system can manage data peaks over maximum transfer rate, ensuring stable communication of variable data flows. The API implements a software watchdog timer to recover from unexpected situations. EFY